RENAL HOMEOSTASIS: KEY PATHWAYS + NEGATIVE FEEDBACK

Solid → response | Red dashed —| negative feedback

OSMOLALITY / WATER

Plasma osmolality ↑

– osmotic stimulus

Osmoreceptors / hypothalamus

VP (ADH) ↑

Thirst ↑

V_2 → AQP2 ↑

Water reabsorption ↑

Drinking ↑

Concentrating support: loop NaCl gradient + VP–UT-A1 urea; vasa recta preserve gradient

Water balance restored / Osmolality ↓

VOLUME / BP

Blood volume / BP ↓

– depletion stimulus

Atrial + arterial stretch ↓

Baroreceptor firing ↓

VP ↑

Sympathetic activity ↑

Water retention ↑

Vasoconstriction ↑

High VP also: V_{1a} → vasoconstriction

SNS also: renal renin release ↑

Volume / BP restored

Stretch / firing recover → VP & SNS drive ↓

RENIN – ANGIOTENSIN

Renal perfusion ↓ / Macula-densa NaCl delivery ↓ / Renal SNS ↑

– renin stimuli

JG cells: Renin ↑

Angiotensin II ↑

FOUR PARALLEL EFFECTS

1 Aldosterone ↑ → Na^+ retention

2 Thirst ↑ → drinking

3 VP ↑ → water retention

4 Vasoconstriction ↑ (AT_1 + central SNS)

Circulating volume / BP restored

POTASSIUM

Plasma K^+ ↑

– K^+ stimulus

Adrenal cortex: Aldosterone ↑

ANGII also stimulates aldosterone (see RAAS pathway)

Distal nephron: MR / ROMK

Urinary K^+ excretion ↑

Plasma K^+ restored

Pathways interact through shared VP, ANGII, aldosterone and thirst. Water concentration, Na^+ content, volume and BP are different variables.